

Practical 3

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Malware Analysis Spring 2025 - P0x03

Executive Summary

From my analysis, I have concluded that sample3.exe is a Darkside ransomware variant.

The file, sample3.exe, not long after it is run, encrypts the victim's files, and the author demands money to be paid for the files to be unencrypted. The malware also changes the background of the victim's computer to a ransom note, which tells them to download a web browser that can access the dark web, and then go to their website on the dark web to pay a ransom. It is also trying to infect other systems associated with the victim through a brute force method of identifying other machines with the same starting 9 digits of the IP address (192.168.245.XXX).

The reason I am sure that this malware is ransomware is that it follows all of the obvious signs of ransomware including encryption of a user's files and a ransom note left for the user to find.

Basic Static Analysis

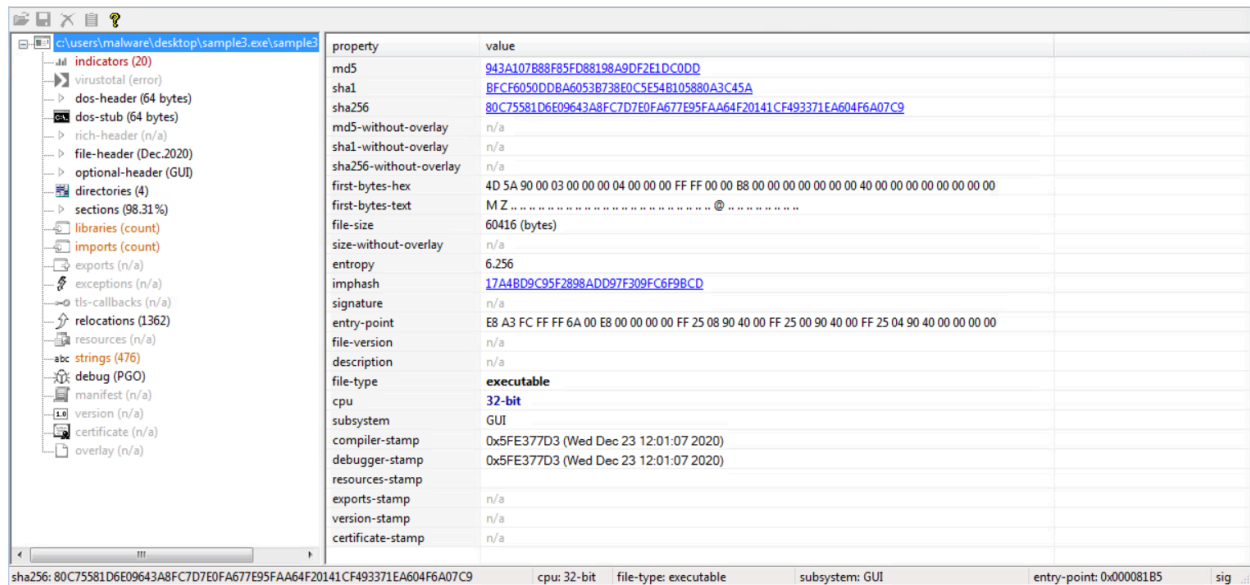


Figure 1: PEStudio Overview

From this screenshot of the apparent compilation date of the malware, which seems to be **December 23, 2020 at 12:01:07 pm**. In addition to this, we can observe that this program is a **Windows GUI program**. It is **32-bit compatible**. Its hashes, as it appears in PE Studio, are 943A107B88F85FD88198A9DF2E1DC0DD (**MD5**) BFCF6050DDBA6053B738E0C5E54B105880A3C45A (**SHA1**) and 80C75581D6E09643A8FC7D7E0FA677E95FAA64F20141 CF493371EA604F6A07C9 (**SHA256**).

property	value	value	value	value	
name	.text	.rdata	.data	.reloc	
md5	52C7889BA37F3177EFE8F5A...	0B8F292A6DD252797194DE2...	C033C139FE22AEF5A9A7633...	BA68B2ECA00E5A89428D63...	
entropy	6.269	3.014	5.673	6.501	
file-ratio (98.31 %)	48.31 %	0.85 %	44.07 %	5.08 %	
raw-address	0x00000400	0x00007600	0x00007800	0x0000E000	
raw-size (59392 bytes)	0x00007200 (29184 bytes)	0x00000200 (512 bytes)	0x00006800 (26624 bytes)	0x00000C00 (3072 bytes)	
virtual-address	0x00401000	0x00409000	0x0040A000	0x00413000	
virtual-size (66177 bytes)	0x000071D3 (29139 bytes)	0x00000176 (374 bytes)	0x0000844C (33868 bytes)	0x00000AEC (2796 bytes)	
entry-point	0x000081B5	-	-	-	
characteristics	0x60000020	0x40000040	0xC0000040	0x42000040	
writable	-	-	x	-	
executable	x	-	-	-	
shareable	-	-	-	-	
discardable	-	-	-	x	
initialized-data	-	x	x	x	
uninitialized-data	-	-	-	-	
unreadable	-	-	-	-	
self-modifying	-	-	-	-	
virtualized	-	-	-	-	
file	n/a	n/a	n/a	n/a	

Figure 2: PESTudio Sections

Packing

From looking at the PESTudio Sections, the .text section contains the entry point and is the only executable section. The .data section is the only writable section. The entropy of all of these sections are below 7, but not out of the realm of possibility of not being packed. When doing further analysis, we can see that the virtual and raw sizes of the data are different but the ratio between the raw and virtual being 98.31% is probably too low for packing. However, this all gets thrown out the window later when we see that there are only 3 imported functions, all of which are used in packing/obfuscation. So this malware is definitely **packed**. More details on this will be in the imports section.

PE File Sections

The PE file sections are named **.txt**, **.rdata**, **.data**, and **.reloc**.

.txt is typically used to store text data like readable strings or instructions, **.rdata** contains constants that shouldn't be modified during execution such as constant variables, string literals, etc. **.data** contains global and static variables that are initialized by the programmer. Lastly, **.reloc** usually holds info to adjust addresses in a program when it is loaded into memory.

PE Function Imports

[illegible]

Figure 3: PEStudio Function Imports

From looking at the function imports, there is a glaring issue. The only functions that are shown are ExitProcess, GetProcAddress, and LoadLibraryA. This means that there is some major obfuscation going on, and the functions this program *actually* calls are going to be done dynamically. This is a classic case of obfuscation and is used by malware authors to hide what is actually going on behind the scenes to deter malware analysts.

Strings

When looking at the **strings**, some strings that seemed important were:

1. **ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/,** which after doing some research, turns out to be the base64 character set, which implies that there is some base64 encoding in this malware.
2. **?456789;,<=,** which seems to be something encoded in base64. Not sure at this point however.
3. **VMProtect begin and VMProtect end,** which are used by malware authors to obfuscate code by turning instructions into virtual instructions, which hides how the malware actually works by letting a virtual machine interpret the code, which means the machine code it reads is not real machine code.

Other files that were imported include: **KERNAL32.dll**, although I'm not sure of their significance at this point.

For this specific piece of malware, there were no visible exports, or an overlay like previous practicals.

Dynamic Analysis

To start, here are the steps I took before starting:

Windows 7:

- 1) Loaded Regshot
- 2) Loaded Process Explorer
- 3) Loaded process monitor, cleared all previous processes

Remnux:

- 1) Loaded FakeDNS
- 2) Loaded Wireshark and began listening

Back to Windows 7:

- 1) Took first shot on Regshot
- 1) Ran the malware (nothing special, just an .exe)
- 2) Waited like a minute then took the second shot and paused the other tools

General Overview of Visible Malware Changes

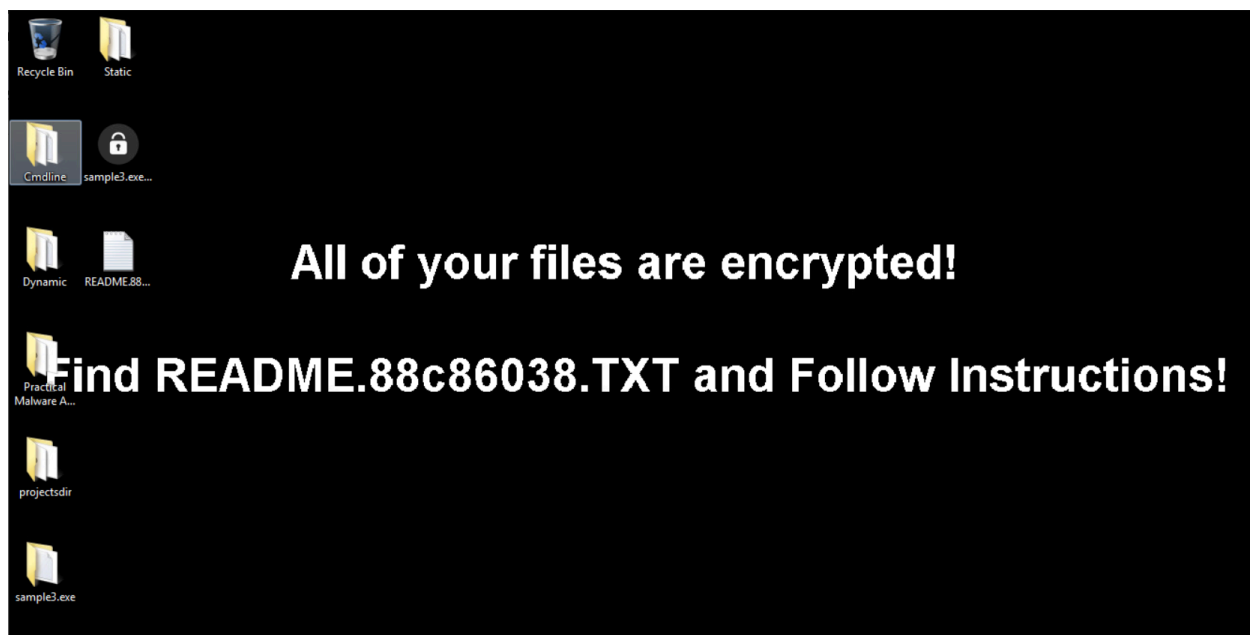
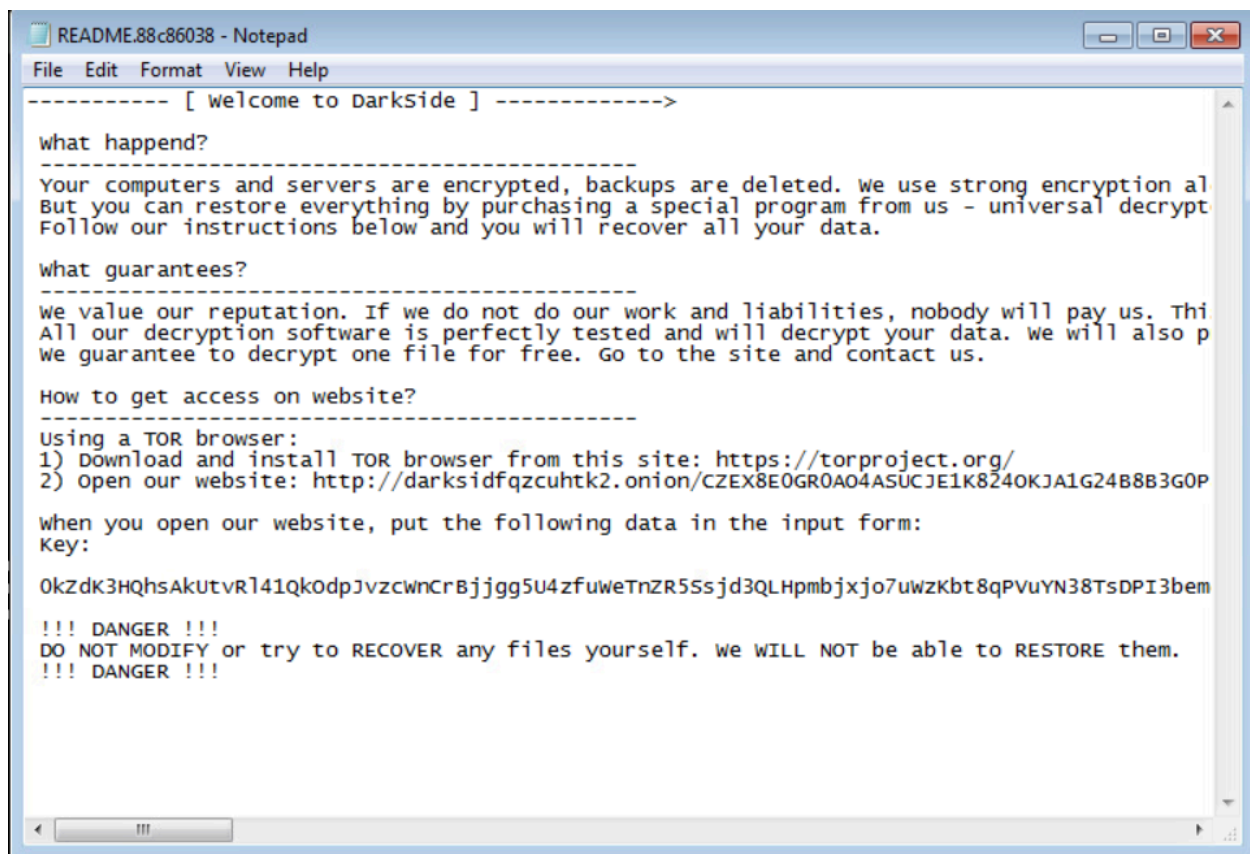


Figure 4: Background with malware message

After running the malware for about 30 seconds, I noticed that my background changed on the Windows 7 computer (the one I ran the malware on). As intended, I then opened the README that had been put into every single folder on the Desktop, as well as conveniently placed right on the Desktop itself.



```
README.88c86038 - Notepad
File Edit Format View Help
----- [ welcome to DarkSide ] ----->

what happend?
-----
Your computers and servers are encrypted, backups are deleted. We use strong encryption al
But you can restore everything by purchasing a special program from us - universal decrypt
Follow our instructions below and you will recover all your data.

what guarantees?
-----
we value our reputation. If we do not do our work and liabilities, nobody will pay us. Thi
All our decryption software is perfectly tested and will decrypt your data. we will also p
we guarantee to decrypt one file for free. Go to the site and contact us.

How to get access on website?
-----
Using a TOR browser:
1) Download and install TOR browser from this site: https://torproject.org/
2) open our website: http://darksidfqzcuhtk2.onion/CZEX8E0GR0A04ASUCJE1K824OKJA1G24B8B3G0P

when you open our website, put the following data in the input form:
Key:
0kZdK3HQhsAkutvRl41QkodpJvzcwnCrBjjgg5U4zfuweTnZR5Ssjd3QLHpmbjxjo7uwzKbt8qPVuYN38TsDPI3bem

!!! DANGER !!!
DO NOT MODIFY or try to RECOVER any files yourself. we WILL NOT be able to RESTORE them.
!!! DANGER !!!
```

Figure 5: README.88c86038.txt

In the README, the malware author describes that all the victim's files have been encrypted, and they would need to install the TOR browser, which allows for dark web searches (torproject(.org)). Then once that is done, they would need to go to a dark web website and pay them directly. The author knows which victim it is by the unique key shown below the instructions.

From the note, I gathered that this was a variation of the Darkside Ransomware that affected Colonial Pipeline on May 8th, 2021. The article describing this attack can be found in [1]. More information specifically about Darkside, the cyberterrorist organization, and the malware they produce can be found in [2].

Created / Modified Registry Keys/Files

```
-----
Keys deleted:39
-----
HKLM\SYSTEM\ControlSet001\services\vmvss
HKLM\SYSTEM\ControlSet001\services\VSS
HKLM\SYSTEM\ControlSet001\services\VSS\Diag
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\SPP
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\SystemRestore
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\ASR Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\COM+ REGDB Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\MSSearch Service Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\Registry Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\Shadow Copy Optimization Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\System Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\VolSnap
HKLM\SYSTEM\ControlSet001\services\VSS\Diag\WMI Writer
HKLM\SYSTEM\ControlSet001\services\VSS\Providers\{b5946137-7b9f-4925-af80-51abd60b20d5}
HKLM\SYSTEM\ControlSet001\services\VSS\Providers\{b5946137-7b9f-4925-af80-51abd60b20d5}\CLSID
HKLM\SYSTEM\ControlSet001\services\VSS\Settings
HKLM\SYSTEM\ControlSet001\services\VSS\Settings\Writers\BlockingRevert
HKLM\SYSTEM\ControlSet001\services\VSS\VssAccessControl
HKLM\SYSTEM\CurrentControlSet\services\vmvss
HKLM\SYSTEM\CurrentControlSet\services\VSS
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\SPP
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\SystemRestore
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\ASR Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\COM+ REGDB Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\MSSearch Service Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\Registry Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\Shadow Copy Optimization Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\System Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\VolSnap
HKLM\SYSTEM\CurrentControlSet\services\VSS\Diag\WMI Writer
HKLM\SYSTEM\CurrentControlSet\services\VSS\Providers
HKLM\SYSTEM\CurrentControlSet\services\VSS\Providers\{b5946137-7b9f-4925-af80-51abd60b20d5}
HKLM\SYSTEM\CurrentControlSet\services\VSS\Providers\{b5946137-7b9f-4925-af80-51abd60b20d5}\CLSID
-----
```

Figure 4: Regshot comparison results from running the malware

```
-----
Values added:67
-----
HKLM\SOFTWARE\Classes\88c86038: "88c86038"
HKLM\SOFTWARE\Classes\88c86038\DefaultIcon: "C:\users\malware\AppData\Local\88c86038.ico"
HKLM\SOFTWARE\Microsoft\WBEM\WMIDREG\SCSI\Disk\ven.VMware\Prod_Virtual_disk\5435015fb0a0&000000_0-[05901221-D566-11d1-82F0-00A0C9062910]: "LowDateTime:803713417,HighD
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto Update\UAS\UpdateCount: 0x00000000
HKU\DEFAULT\Software\Classes\Local Settings\MuiCache\2752c6487e\BC:\Program Files\Common Files\System\wab32res.dll,-10100: "Contacts"
HKU\DEFAULT\Software\Classes\Local Settings\MuiCache\2752c6487e\BC:\Windows\System32\ieframe.dll,-12385: "Favorites Bar"
HKU\DEFAULT\Software\Classes\Local Settings\MuiCache\2752c6487e\BC:\Windows\System32\notepad.exe,-469: "Text document"
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\txt\OpenWithList\{a: "NOTEPAD.EXE"
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\FileExts\txt\OpenWithList\MRUList: "a"
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\{52 00 45 00 41 00 44 00 40 00 45 00 2E 00 38 00 0
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\TXT\0: 52 00 45 00 41 00 44 00 40 00 45 00 2E 00 38 00 3
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\TXT\MRULISTEX: 00 00 00 00 FF FF FF FF
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count\{06523100-02
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count\p\Hrref.zny
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\S.O\Cache\Extensible Cache\MSHist012025041520250416\CachePri
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\S.O\Cache\Extensible Cache\MSHist012025041520250416\CachePat
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\S.O\Cache\Extensible Cache\MSHist012025041520250416\CacheKey
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\S.O\Cache\Extensible Cache\MSHist012025041520250416\CacheLi
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\00-50-56-87-82-ca\wpadDecisionReason: 0x00000001
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\00-50-56-87-82-ca\wpadDecisionTime: F0 06 8E 33 79 AE I
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\00-50-56-87-82-ca\wpadDecision: 0x00000000
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\{A28CA5F8-40FA-4D1C-9DE7-37CB80ECD3C8}\wpadNetworkName
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\{A28CA5F8-40FA-4D1C-9DE7-37CB80ECD3C8}\wpadDecisionTime
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\{A28CA5F8-40FA-4D1C-9DE7-37CB80ECD3C8}\wpadDecision: 0
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\wpad\{A28CA5F8-40FA-4D1C-9DE7-37CB80ECD3C8}\wpadNetworkName
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\MuiCache\2752c6487e\@SystemRoot\system32\cmstpui.dll,-100: "Connection Manager"
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\RU\12: 5E 00 31 00 00 00 00 90 5A 14 15 10 00 7
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\RU\12\NodesTot: 0x00000049
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\RU\12\MRULISTEX: FF FF FF FF
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\73\Shell\{5C4F28B5-F869-4E84-8E60-F11DB97C5CC7}\R
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\73\Shell\{5C4F28B5-F869-4E84-8E60-F11DB97C5CC7}\FI
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\73\Shell\{5C4F28B5-F869-4E84-8E60-F11DB97C5CC7}\V
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\73\Shell\{5C4F28B5-F869-4E84-8E60-F11DB97C5CC7}\M
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Classes\Local Settings\Software\Microsoft\Windows\Shell\Bags\73\Shell\{5C4F28B5-F869-4E84-8E60-F11DB97C5CC7}\L
-----
```

Figure 5: Regshot comparison (part 2) results from running the malware

Figure 6: Regshot comparison (part 3) results from running the malware

Figure 7: Regshot comparison (part 4) results from running the malware

As for other interesting finds, I saw that WPAD keys were again tampered with, although I think this may be to infect other systems with the malware on the same network, rather than sending data back to the attacker. This fits with the nature of a ransomware attack, which is less about exfiltration and more about encryption of local files.

In addition to that, two changes that I saw that actually changed files/systems that I could see were the changing of the wallpaper through the ControlPanel\Desktop\Wallpaper to an image matching the README (88c86o38.BMP), and the notepad being tampered with, likely doing with the README file setup.

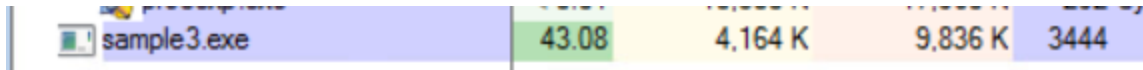


Figure 8: Process Explorer

When viewing the process in Process Explorer after running the malware, there doesn't seem to be any persistence of the malware. It most likely just does its encrypting job and then ends after about 30 seconds. It does run more processes, but they go by in a flash, so I cannot list them here.

Since Process Explorer didn't seem to give me any insight, I moved on to Process Monitor. In Process Monitor, I wanted to see what sample3.exe was actually doing behind the scenes, and when I did this, over three hundred thousand processes were run by this executable. I will highlight the ones I found that seemed interesting, as there are probably too many to fit into this report.

10:42:...	sample3.exe	576	RegCloseKey	HKLM\SOFTWARE\Microsoft\Rpc	SUCCESS	
10:42:...	sample3.exe	576	RegOpenKey	HKLM\System\CurrentControlSet\Control\ComputerName\ActiveComputerName	REPARSE	Desired Access: R...
10:42:...	sample3.exe	576	RegOpenKey	HKLM\System\CurrentControlSet\Control\ComputerName\ActiveComputerName	SUCCESS	Desired Access: R...
10:42:...	sample3.exe	576	RegQueryValue	HKLM\System\CurrentControlSet\Control\ComputerName\ActiveComputerName\ComputerName	SUCCESS	Type: REG_SZ, Le...
10:42:...	sample3.exe	576	RegCloseKey	HKLM\System\CurrentControlSet\Control\ComputerName\ActiveComputerName	SUCCESS	

Figure 9: Process Monitor

The first of these interesting processes was a registry opening of ActiveComputerName. I would assume this is used by the malware author to get the victim's name.

10:42:...	sample3.exe	576	RegOpenKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE	SUCCESS	Desired Access: Q...
10:42:...	sample3.exe	576	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE\sample3.exe	NAME NOT FOUND	Length: 144
10:42:...	sample3.exe	576	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE\	NAME NOT FOUND	Length: 144
10:42:...	sample3.exe	576	RegCloseKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE	SUCCESS	

Figure 10: Process Monitor

Next, I found the malware tampers with the "FEATURE_HTTP_USERNAME_PASSWORD_DISABLE" registry key, which could be to clear or prevent auto-login credentials from being reused.

10:42:...	sample3.exe	3140	CreateFile	C:\Users\malware\Desktop\sample3.exe\CRYPTBASE.dll	NAME NOT FOUND	Desired Access: R...
10:42:...	sample3.exe	3140	CreateFile	C:\Windows\System32\cryptbase.dll	SUCCESS	Desired Access: R...
10:42:...	sample3.exe	3140	CreateFile	C:\Windows\System32\cryptbase.dll	SUCCESS	Desired Access: R...

Figure 11: Process Monitor

Since I know this malware is ransomware that encrypts a victim's files, I made sure to look for CRYPTBASE.dll, which can be abused by malware authors to do just that. Sample3, of course, is observed using this file.

2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\Prefetch\SAMPLE3.EXE-6...	NAME NOT FOUND	Desired Access: G...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.exe	SUCCESS	Desired Access: E...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\sechost.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\sechost.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\imm32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.ex...	NAME NOT FOUND	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\mpr.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\mpr.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.ex...	NAME NOT FOUND	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\IPHLPAPI.DLL	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\IPHLPAPI.DLL	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.ex...	NAME NOT FOUND	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\winnsi.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\winnsi.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.ex...	NAME NOT FOUND	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\netapi32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\netapi32.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Users\malware\Desktop\sample3.ex...	NAME NOT FOUND	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\netutils.dll	SUCCESS	Desired Access: R...
2:21:1...	sample3.exe	3520	CreateFile	C:\Windows\System32\netutils.dll	SUCCESS	Desired Access: R...

Figure 12: Process Monitor

I also looked into what files this malware accesses, and honestly, there are well over 20 .dll files it accesses including **imm32.dll**, **mpr.dll**, and more.

9	1.072808563	192.168.245.130	192.168.245.133	DNS	85 Standard query 0x7298 A teredo.ipv6.microsoft.com
10	1.073343035	192.168.245.133	192.168.245.130	DNS	101 Standard query response 0x7298 A teredo.ipv6.microsoft.com A 192.168.245.133
13	2.422789363	192.168.245.129	192.168.245.133	DNS	92 Standard query 0x0559 A tlu.dl.delivery.mp.microsoft.com
14	2.423346748	192.168.245.133	192.168.245.129	DNS	108 Standard query response 0x0559 A tlu.dl.delivery.mp.microsoft.com A 192.168.245...
17	2.900222970	192.168.245.129	192.168.245.133	DNS	89 Standard query 0x4abe A v10.events.data.microsoft.com
18	2.900657284	192.168.245.133	192.168.245.129	DNS	105 Standard query response 0x4abe A v10.events.data.microsoft.com A 192.168.245.133
40	4.917579735	192.168.245.130	192.168.245.133	DNS	79 Standard query 0x1872 A securebestapp20.com
41	4.918131027	192.168.245.133	192.168.245.130	DNS	95 Standard query response 0x1872 A securebestapp20.com A 192.168.245.133
229	42.742680053	192.168.245.129	192.168.245.133	DNS	86 Standard query 0x6533 A slscr.update.microsoft.com
230	42.743163338	192.168.245.133	192.168.245.129	DNS	102 Standard query response 0x6533 A slscr.update.microsoft.com A 192.168.245.133
233	42.843500976	192.168.245.129	192.168.245.133	DNS	91 Standard query 0x04af A settings-win.data.microsoft.com
234	42.843928637	192.168.245.133	192.168.245.129	DNS	107 Standard query response 0x04af A settings-win.data.microsoft.com A 192.168.245...
243	43.341790518	192.168.245.129	192.168.245.133	DNS	92 Standard query 0xb01f A watson.events.data.microsoft.com
244	43.342401032	192.168.245.133	192.168.245.129	DNS	108 Standard query response 0xb01f A watson.events.data.microsoft.com A 192.168.245...
475	47.359330841	192.168.245.130	192.168.245.133	DNS	86 Standard query 0x74d1 PTR 5.245.168.192.in-addr.arpa
476	47.359843472	192.168.245.133	192.168.245.130	DNS	102 Standard query response 0x74d1 PTR 5.245.168.192.in-addr.arpa A 192.168.245.133
905	50.540856016	192.168.245.130	192.168.245.133	DNS	88 Standard query 0xf9c0 PTR 128.245.168.192.in-addr.arpa
906	50.541043818	192.168.245.130	192.168.245.133	DNS	88 Standard query 0x8bf1 PTR 129.245.168.192.in-addr.arpa
907	50.541392932	192.168.245.133	192.168.245.130	DNS	104 Standard query response 0xf9c0 PTR 128.245.168.192.in-addr.arpa A 192.168.245.1...
908	50.541433609	192.168.245.130	192.168.245.133	DNS	88 Standard query 0x4b8c PTR 133.245.168.192.in-addr.arpa
909	50.541644644	192.168.245.133	192.168.245.130	DNS	104 Standard query response 0x8bf1 PTR 129.245.168.192.in-addr.arpa A 192.168.245.1...
910	50.541845861	192.168.245.133	192.168.245.130	DNS	104 Standard query response 0x4b8c PTR 133.245.168.192.in-addr.arpa A 192.168.245.1...
1193	53.713697783	192.168.245.130	192.168.245.133	DNS	86 Standard query 0x886f A 5.245.168.192.in-addr.arpa

Figure 13: Wireshark capturing DNS results from running the malware

Running the malware resulted in one interesting DNS request from **securebestapp20(.com)**. I'm not sure of the significance at this point.

```
fakedns[INFO]: Response: securebestapp20.com -> 192.168.245.133
```

Figure 14: FakeDNS capturing DNS results from running the malware

I confirmed this suspicion with FakeDNS.

434	73.010614446	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.2? Tell 192.168.245.130
435	73.010614506	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.3? Tell 192.168.245.130
436	73.010614556	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.4? Tell 192.168.245.130
437	73.010614617	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.6? Tell 192.168.245.130
438	73.010614667	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.7? Tell 192.168.245.130
439	73.010614717	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.8? Tell 192.168.245.130
440	73.010845951	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.9? Tell 192.168.245.130
441	73.010846051	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.10? Tell 192.168.245.130
442	73.010846101	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.11? Tell 192.168.245.130
443	73.010846161	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.12? Tell 192.168.245.130
444	73.010846211	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.13? Tell 192.168.245.130
445	73.010846261	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.14? Tell 192.168.245.130
446	73.010846322	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.15? Tell 192.168.245.130
447	73.010846372	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.16? Tell 192.168.245.130
448	73.010858104	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.17? Tell 192.168.245.130
449	73.010858164	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.18? Tell 192.168.245.130
450	73.010858224	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.19? Tell 192.168.245.130
451	73.010858284	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.20? Tell 192.168.245.130
452	73.010858334	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.21? Tell 192.168.245.130
453	73.010858394	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.22? Tell 192.168.245.130
454	73.010858444	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.23? Tell 192.168.245.130
455	73.010858504	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.24? Tell 192.168.245.130
456	73.010871248	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.25? Tell 192.168.245.130
457	73.010871309	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.26? Tell 192.168.245.130
458	73.010871369	00:50:56:87:c0:26	Broadcast	ARP	60 Who has 192.168.245.27? Tell 192.168.245.130

Figure 15: Wireshark capturing ARP results from running the malware

Looping back to Wireshark, I saw that the Windows 7 machine was looking for what other systems are on the same network. It is trying to infect other systems associated with the victim through a brute force method of identifying other machines with the same starting 9 digits of the IP address (192.168.245.XXX). The ransom note *does* say computers and servers *plural*, so this makes sense. ARP connection is commonly used by malware authors to infect other computers that are connected to each other. These kinds of requests are great for finding out what other computers are on the network, without the need for admin privileges.

Scylla / x32dbg Dynamic Analysis

Since this malware seems to be hiding which functions it is abusing before runtime, I figured it best to try and dump the contents right after GetProcAddress was executed in the malware. This reaped some beneficial results.

c:\users\malware\desktop\sample1_dump_scylla	property	value
indicators (41)	md5	BE9B96C165A66B1E785042F279970DC7
virustotal (error)	sha1	1D4A444407988D14115CF66ACBC44DCD55031B6C
dos-header (64 bytes)	sha256	8E6DBA45120D5212D35F2C99C190725C65693AE9AA1E4C047736FC11152094F5
dos-stub (64 bytes)	md5-without-overlay	n/a
rich-header (n/a)	sha1-without-overlay	n/a
file-header (Dec.2020)	sha256-without-overlay	n/a
optional-header (GUI)	first-bytes-hex	4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 88 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
directories (3)	first-bytes-text	M Z@.....
sections (self-modifying)	file-size	86528 (bytes)
libraries (3)	size-without-overlay	n/a
imports (71)	entropy	6.412
exports (n/a)	imphash	3A687B787D183F251E1E48AE70A52F23
exceptions (n/a)	signature	n/a
tls-callbacks (n/a)	entry-point	E8 A3 FC FF FF 6A 00 E8 00 00 00 00 FF 25 08 90 1D 01 FF 25 04 90 1D 01 00 00 00
relocations (1362)	file-version	n/a
resources (n/a)	description	n/a
strings (1389)	file-type	executable
debug (PGO)	cpu	32-bit
manifest (n/a)	subsystem	GUI
version (n/a)	compiler-stamp	0x5FE377D3 (Wed Dec 23 12:01:07 2020)
certificate (n/a)	debugger-stamp	0x5FE377D3 (Wed Dec 23 12:01:07 2020)
overlay (n/a)	resources-stamp	n/a
	exports-stamp	n/a
	version-stamp	n/a
	certificate-stamp	n/a

Figure 16: PEStudio Overview after dumping the .exe with Scylla

name (71)	group (3)	type (1)	ordinal (0)	blacklist (4)	anti-debug (0)	undocumented (0)	deprecated (4)	library (3)
RtlCaptureContext	exception-handling	implicit	-	x	-	-	-	ntdll.dll
RtlCaptureStackBackTrace	diagnostic	implicit	-	x	-	-	-	ntdll.dll
EnumCalendarInfoW	-	implicit	-	x	-	-	-	kernelbase.dll
SetFileApisToOEM	-	implicit	-	x	-	-	-	kernelbase.dll
EnumSystemLocalesEx	system-information	implicit	-	-	-	-	-	kernelbase.dll
EnumSystemLocalesW	system-information	implicit	-	-	-	-	-	kernelbase.dll
GetStringTypeA	memory	implicit	-	-	-	-	x	kernelbase.dll
VirtualAllocExNuma	memory	implicit	-	-	-	-	-	kernelbase.dll
RtlUnwind	-	implicit	-	-	-	-	-	ntdll.dll
InitOnceInitialize	-	implicit	-	-	-	-	-	kernel32.dll
BaseReleaseProcessDllPath	-	implicit	-	-	-	-	-	kernelbase.dll
BaseGetProcessDllPath	-	implicit	-	-	-	-	-	kernelbase.dll
BaseGetProcessDllPath	-	implicit	-	-	-	-	-	kernelbase.dll
LoadStringByReference	-	implicit	-	-	-	-	x	kernelbase.dll
InternalCidToName	-	implicit	-	-	-	-	-	kernelbase.dll
NtUserDefaultLocale	-	implicit	-	-	-	-	-	kernelbase.dll
GetUserInfo	-	implicit	-	-	-	-	-	kernelbase.dll
GetPtrCalDataArray	-	implicit	-	-	-	-	-	kernelbase.dll
GetPtrCalData	-	implicit	-	-	-	-	-	kernelbase.dll
GetStringTableEntry	-	implicit	-	-	-	-	-	kernelbase.dll
CheckGroupPolicyEnabled	-	implicit	-	-	-	-	-	kernelbase.dll
OpenRegKey	-	implicit	-	-	-	-	-	kernelbase.dll
GetCPHashNode	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumSystemCo...	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumUllanguag...	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumUllanguag...	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumSystemlan...	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumDateForm...	-	implicit	-	-	-	-	-	kernelbase.dll
Internal_EnumTimeForm...	-	implicit	-	-	-	-	-	kernelbase.dll
NtValidateLocale	-	implicit	-	-	-	-	-	kernelbase.dll

Figure 17: PESTudio Imports after dumping the .exe with Scylla

Unlike the previous imports section with just the 3 functions, this Scylla dumped version has 71 imports. This seems to be a fraction of the actual functions that are utilized by this malware. I could not find a way to unpack the malware and find ALL of the functions it uses.

Ghidra Analysis

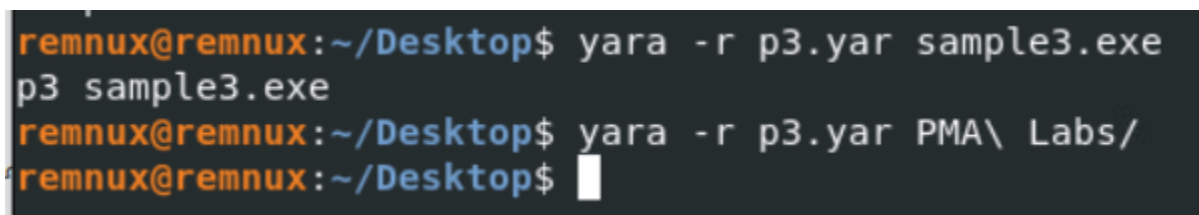
```
Decompile: FUN_00008996 - (sample3.exe)
2 int FUN_00008996(undefined4 *param_1,int param_2,undefined4 *param_3)
3
4 {
5     undefined4 *puVar1;
6     ushort uVar2;
7     byte bVar3;
8     ushort uVar4;
9     uint uVar5;
10    undefined4 *puVar6;
11
12    puVar6 = param_3;
13    do {
14        if (param_2 == 0) goto LAB_00008a14;
15        uVar2 = CONCAT11(*param_1,*param_1) & 0x3ff;
16        uVar4 = CONCAT11(param_1[1],param_1[1]) & 0xfff;
17        uVar5 = CONCAT11(param_1[2],param_1[2]) & 0xffff3fff;
18        bVar3 = (char)(uVar2 >> 8) << 4 | (byte)uVar4 >> 4;
19        param_1 = param_1 + 3;
20        puVar1 = puVar6 + 1;
21        *puVar6 = CONCAT22(CONCAT11(s_ABCDEFGHIJKLMNOPQRSTUVWXYZabcdef_00017ce6[uVar5 >> 8],
22                                s_ABCDEFGHIJKLMNOPQRSTUVWXYZabcdef_00017ce6
23                                [(byte)((byte)uVar5 >> 6 | (char)(uVar4 >> 8) << 2)]),
24                                CONCAT11(s_ABCDEFGHIJKLMNOPQRSTUVWXYZabcdef_00017ce6[bVar3],
25                                s_ABCDEFGHIJKLMNOPQRSTUVWXYZabcdef_00017ce6
26                                [CONCAT11(bVar3,(byte)uVar2 >> 2) & 0xff]));
27        param_2 = param_2 + -3;
28        puVar6 = puVar1;
29    } while (-1 < param_2);
30    puVar6 = (undefined4 *)((int)puVar1 + param_2);
31    do {
32        *(undefined4 *)puVar6 = 0x3d;
33        puVar6 = (undefined4 *)((int)puVar6 + 1);
34        param_2 = param_2 + 1;
35    } while (param_2 != 0);
36 LAB_00008a14:
```

Figure 18: Suspicious function in Ghidra

In Ghidra, I mainly wanted to look for where the malware encrypts all of the files. This function, FUN_00008996, is the most likely to be in charge of part of the encryption. This encryption is a base64 encryption, indicated by the “ABCDEFGHIJKLMNOPQRSTUVWXYZ...” string located throughout the program (which is the base64 alphabet).

Indicators of Compromise

The main network-based indicators I observed were the suspicious DNS request to visit `securebestapp20(.com)`, and the ARP requests trying to find other systems for the malware to infect. Moving on to the more obvious signs a user is infected are the changing of the wallpaper to the ransom message, the ransom .txt file (*README.88c86038.txt*) that is placed in many of the folders on the victim's computer, and the obvious encryption of most of the users files, although it was odd that the malware did not encrypt many of the malware analysis tools.

A terminal window with a dark background and light-colored text. The prompt is 'remnux@remnux:~/Desktop\$'. The first command is 'yara -r p3.yar sample3.exe', followed by the output 'p3 sample3.exe'. The second command is 'yara -r p3.yar PMA\ Labs/', followed by the output 'remnux@remnux:~/Desktop\$' on a new line, indicating no match was found.

```
remnux@remnux:~/Desktop$ yara -r p3.yar sample3.exe
p3 sample3.exe
remnux@remnux:~/Desktop$ yara -r p3.yar PMA\ Labs/
remnux@remnux:~/Desktop$
```

Figure 19: Yara rule successfully identifying sample3.exe and not anything from the PMA Labs

Here is the YARA rule I ended up using to identify sample3.exe:

```
rule p3
{
    strings:
        $base64alphabet =
        "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/"
        $specific64string = "?456789:;<="

    condition:
        $base64alphabet and $specific64string
}
```

Appendix

[1] Freed, A. M. (n.d.). *Inside the darkside ransomware attack on Colonial Pipeline.* Cybersecurity Software.

<https://www.cybereason.com/blog/inside-the-darkside-ransomware-attack-on-colonial-pipeline#:~:text=On%20May%208%2C%20the%20Colonial,ransomware%20attack%20a%20day%20earlier.>

[2] *What is darkside ransomware?* | Akamai. (n.d.).

<https://www.akamai.com/glossary/what-is-darkside-ransomware>